

# Tuhin Garai

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## SKILLS

**Programming/Scripting:** Python, Bash, Shell Scripting

**DevOps & Cloud:** Docker, Docker Compose, Kubernetes, Jenkins, CI/CD, AWS (EC2, VPC, S3, IAM, SNS, CloudWatch), GitHub Actions, Ansible, Terraform

**Tools & Platforms:** Git, GitHub, Linux

**Systems & Administration:** Linux Administration, System Monitoring, Troubleshooting, Log Analysis

## PROJECTS

**Yet Another LuCI App**, Android Application | In Development

- Developing a native Android client for OpenWrt administration using ubus/RPCD, supporting package management, firewall control, network topology, wireless clients, and multi-router management.
- Implemented secure on-device credential storage using hardware-backed encryption and designed workflows for managing real OpenWrt environments from mobile.
- Published a tester-only Google Play pre-release and open-source community edition on GitHub.

**Zero Downtime Application Deployment using Kubernetes & AWS**

- Designed a highly available cloud-native deployment architecture on Amazon EKS with multi-region failover using Route 53 health checks.
- Implemented Kubernetes rolling updates, readiness/liveness probes, and Horizontal Pod Autoscaler (HPA) for achieving zero-downtime deployments and self-healing recovery across simulated high-traffic scenarios.
- Integrated Prometheus and Grafana dashboards alongside CloudWatch for end-to-end observability of system health and scaling behavior.
- Configured warm standby disaster recovery with automated failover, reducing projected RTO by ~40% compared to cold standby baseline.

**Serverless EC2 Monitor & Control Dashboard**

- Built a serverless EC2 monitoring and automation system using AWS Lambda, API Gateway, DynamoDB, and S3; published findings in IJRSET.
- Automated EC2 instance lifecycle management using usage-based triggers, reducing manual monitoring and shutdown effort by ~1 hour daily.
- Configured AWS-based monitoring, alerting, and scalable infrastructure workflows to improve observability, reliability, response time, and operational availability.
- Documented deployment architecture, workflows, and operational processes.

## ACHIEVEMENTS

**Android Open Source (AOSP) Community Contributor**

- Tested and validated custom Android ROMs (crDroid, LineageOS, AOSP, VoltageOS, and others) across 5+ devices including Nothing Phone 2, Xiaomi, and Poco series.
- Identified and reported bugs, system instability, and performance bottlenecks; captured diagnostic logs (logcat) to assist developers in debugging and optimization.
- Performed device flashing, recovery, and restoration via fastboot, TWRP, OrangeFox, and EDL mode across multiple device architectures.
- Worked with Magisk, KernelSU, and LSPosed for root configuration and system-level customization testing.

## PUBLICATIONS

**Research Paper: IJRSET — Build a Serverless EC2 Monitor & Control Dashboard**

## EDUCATION

**Bachelors in Computer Application (BCA)**, Chandigarh University

2021 – 2024 | Mohali, Punjab

**Masters in Computer Application (MCA)**, Jain University

2024 – 2026 | Bengaluru, Karnataka

Specialization: Storage & Cloud Technology (Major), AI & ML (Minor)